

Six-Point One-Hole Compression for Five-Cycle Double Covers

Atharva Vaidya

AI-assisted research draft for independent human verification

July 29, 2026

Resolution status and trust boundary. The standard five-cycle double cover conjecture remains open. This manuscript proves an exact affine characterization after a nowhere-zero $\mathbb{F}_2^3$ -flow, an omitted coordinate pair, and a merge pair have been fixed. Existentially quantifying those choices gives exactly the original conjecture, not a proof of it. The orientable conjecture is not considered.

OpenAI Codex agents, under Atharva Vaidya’s direction, discovered the normal form, derived the proof and local tables, wrote both computational implementations, performed the enumerations, and drafted this manuscript. The second implementation is independent of the first implementation, not independent of AI. No human peer review is claimed. A human author must check the proof, rerun or reimplement the finite audits, review prior art, and accept scholarly responsibility before submission.

Abstract

Let $W = \mathbb{F}_2^3$, let G be a finite loopless cubic multigraph, and fix a nowhere-zero W -flow f . The standard eight-coordinate triangle construction labels every edge by a two-subset of W . We show that this construction compresses to a five-cycle double cover if its labels use only six points and omit one pair among them: identify the two points of that unused pair.

For fixed disjoint pairs $R, M \in \binom{W}{2}$, where R is omitted from the support and M is the pair to be identified, existence of such a labelling is exactly an affine system over $\mathbb{F}_2$, with three potential bits per graph vertex. We give a complete human proof and classify every local list. Up to affine equivalence, R and M are parallel or skew; the seven local list sizes are respectively

$$(4, 2, 2, 2, 2, 2, 2) \quad \text{and} \quad (4, 4, 2, 2, 2, 1, 1).$$

Failure for a fixed triple (f, R, M) has the elementary dual certificate $y^\top A = 0$, $y^\top b = 1$.

Two independently written exact checkers verify the local classification and a strict 34-vertex fixed-flow example: all 56 five-point restrictions fail, while eight of the 420 six-point one-hole systems succeed. A complete 900-orbit flow census on one rigid 12-vertex graph separates the five-point, six-point one-hole, and unrestricted five-colourable coordinate-potential tests. These are structural and computational results about fixed flows. They neither prove nor refute FiveCDC.

1 Definitions and exact scope

A *cycle* in this manuscript is an Eulerian edge-subset: every vertex has even degree in it. It need not be connected or 2-regular. A *five-cycle double cover* is an indexed family $(C_1, \dots, C_5)$ of such subsets in which every edge occurs exactly twice. Empty members are allowed. Thus “five” means

“at most five,” as in the standard formulation used by Oum [5]. Parallel edges are distinct edge-set elements. We exclude loops here and make no claim about the orientable variant.

The standard conjecture, attributed independently to Celmins and Preissmann, asserts that every finite bridgeless graph has a five-cycle double cover [1, 6, 2, 5]. It remains open. Hušek and Šámal recently gave an equivalent flow-selection condition for cubic graphs [2, Theorem 3.16 and Conjecture 3.19]. Our result is another exact fixed-flow normal form. Its graph-level existential closure is again FiveCDC, so the remaining quantifier cannot be omitted.

There is important earlier configuration-colouring literature. Král’, Máčajová, Pangrác, Raspaud, Sereni, and Škoviera developed projective, affine, and abelian colourings of cubic graphs, including Desargues-configuration equivalences and the F_i/A_i colouring framework [4]. We claim no novelty for those configuration or colouring equivalences. The possible new contribution here is narrower: after fixing a Fano flow, we impose a six-point support and one missing duad, derive its affine potential system and complete local orbit table, and certify a strict separation from the five-point restriction for one fixed flow. Hušek–Šámal are also the prior source used here for the eight-coordinate system and its five-set/component-parity specialization [2].

Put $W = \mathbb{F}_2^3$, written additively. A map

$$f : E(G) \longrightarrow W - \{0\}$$

is a nowhere-zero W -flow when

$$\sum_{e \ni v} f(e) = 0 \quad (v \in V(G)).$$

At a cubic vertex the three incident values are the nonzero elements of a unique two-dimensional linear subspace $H_v \leq W$.

Definition 1.1. An *eight-coordinate pair labelling* is a map

$$P : E(G) \longrightarrow \binom{W}{2}$$

such that, for every $v \in V(G)$ and $z \in W$, the number of incident edges $e \ni v$ with $z \in P_e$ is even. It is *compatible with* f when the xor of the two members of P_e is $f(e)$ for every edge e .

Every coordinate z defines the Eulerian set $\{e : z \in P_e\}$, and every edge belongs to exactly two coordinates. Thus a pair labelling is an eight-cycle double cover with named coordinates. We next parameterize all labellings compatible with a fixed flow.

2 Coordinate triangles and vertex potentials

For a two-dimensional subspace $H \leq W$ and $t \in W$, define

$$\Delta_H(t) = (t + H) - \{t\}. \quad (2.1)$$

This is a three-point affine triangle. Its three sides have differences equal to the three nonzero elements of H .

For an incidence $e \ni v$, put $p = f(e)$, and let $s_{v,e}$ be either one of the other two flow values at v . The side of $\Delta_{H_v}(t_v)$ having difference p is

$$P_{v,e} = t_v + s_{v,e} + \langle p \rangle = \{t_v + s_{v,e}, t_v + s_{v,e} + p\}. \quad (2.2)$$

The two possible choices of $s_{v,e}$ differ by p , so the unordered pair in (2.2) is well defined.

Lemma 2.1 (potential parameterization). *For a fixed nowhere-zero flow f , compatible pair labellings are in bijection with vertex potentials $(t_v : v \in V(G))$ satisfying*

$$t_u + t_v + s_{u,e} + s_{v,e} \in \langle f(e) \rangle \quad (e = uv). \quad (2.3)$$

The labelling is then given by (2.2).

Proof. The two endpoint formulae in (2.2) describe the same unordered pair exactly when their coset representatives differ by an element of $\langle f(e) \rangle$, which is (2.3). At a vertex, the three pairs are the three sides of $\Delta_{H_v}(t_v)$, so each of its three coordinate points occurs twice and every other point occurs zero times. The coordinate parity condition follows.

Conversely, at a cubic vertex, the three labels of any pair labelling form a triangle. Indeed, if the first two pairs are A, B , parity forces the third to be $A \triangle B$; since it has size two, $|A \cap B| = 1$. Write the resulting triangle points as x, y, z . Its edge differences are the three incident flow values, and

$$t_v = x + y + z$$

is the unique potential for which its point set is $\Delta_{H_v}(t_v)$. Agreement of each edge label at its endpoints then gives (2.3). The two constructions are inverse. $\square$

For any nonzero $p \in W$, membership in $\langle p \rangle$ is equivalent to two scalar affine equations: apply a basis of the two-dimensional annihilator of p in W^* . Hence all edge conditions in (2.3) are affine binary equations.

3 The six-point one-hole system

Choose disjoint pairs

$$R, M \in \binom{W}{2}, \quad R \cap M = \emptyset. \quad (3.1)$$

The points of R are forbidden, so the allowed support is $S = W - R$, of size six. The pair $M \subset S$ is also forbidden as an edge label. For a Fano plane H , define

$$\mathcal{A}_H(R, M) = \{t \in W : \Delta_H(t) \subseteq W - R, M \not\subseteq \Delta_H(t)\}. \quad (3.2)$$

The second condition says precisely that M is not one of the three sides of the local triangle.

Theorem 3.1 (affine six-point one-hole criterion). *Fix G, f, R, M as above. There is a compatible pair labelling supported on $W - R$ and not using the pair M if and only if the following system is soluble:*

$$t_v \in \mathcal{A}_{H_v}(R, M) \quad (v \in V(G)), \quad (3.3)$$

$$t_u + t_v + s_{u,e} + s_{v,e} \in \langle f(e) \rangle \quad (e = uv). \quad (3.4)$$

Every set $\mathcal{A}_H(R, M)$ is a nonempty affine subspace of W , of dimension zero, one, or two. Consequently (3.3)–(3.4) is an ordinary affine system over $\mathbb{F}_2$.

Proof. By Lemma 2.1, condition (3.4) is equivalent to endpoint compatibility. The definition (3.2) is exactly the support and missing-pair condition at each vertex. It remains only to prove that every local list is affine and nonempty. This follows from the exhaustive two-case calculation in Theorem 4.1 below. $\square$

To turn a local membership condition into scalar equations, write its nonempty affine subspace as $t_0 + U$. Then

$$t \in t_0 + U \iff \lambda(t) = \lambda(t_0) \text{ for every } \lambda \in U^\perp.$$

Together with the two annihilator equations per edge, this gives a literal matrix $Ax = b$ with three bits per vertex.

4 Complete local classification

Translate the eight coordinate points so that

$$R = \{0, d\}, \quad M = \{a, b\}, \quad m = a + b. \quad (4.1)$$

Translation changes neither pair differences nor the planes H . There are two affine orbits:

- *parallel*, when $m = d$; then $R \cup M = H_0 := \langle d, a \rangle$;
- *skew*, when $m \neq d$; disjointness implies that d, m, a form a basis of W .

Theorem 4.1 (local list table). *In the parallel case,*

$$\mathcal{A}_H(R, M) = \begin{cases} W - H, & H = H_0, \\ M, & d \in H, H \neq H_0, \\ R, & d \notin H. \end{cases} \quad (4.2)$$

Across the seven planes, the list sizes are

$$(4, 2, 2, 2, 2, 2, 2).$$

In the skew case,

$$\mathcal{A}_H(R, M) = \begin{cases} M, & d, m \in H, \\ W - H, & d \in H, m \notin H, \\ \{r \in R : a + r \notin H\}, & d \notin H, m \in H, \\ R, & d, m \notin H. \end{cases} \quad (4.3)$$

Across the seven planes, the list sizes are

$$(4, 4, 2, 2, 2, 1, 1).$$

In particular, every local list is a point, affine line, or affine plane.

Proof. First ignore the restriction on M . If $d \in H$, the forbidden points $0, d$ lie in one H -coset. The triangle must therefore use the other coset, and its omitted point t may initially be any of the four points of that coset. If $d \notin H$, the forbidden points lie in opposite H -cosets, and the only possible omitted points are $t = 0, d$.

Now impose the missing-pair condition. If $m \notin H$, the two members of M lie in opposite H -cosets, so they cannot both occur in one triangle. If $m \in H$, they lie in one coset. When that coset is the chosen one, one of a, b must itself be the omitted point.

In the parallel case $m = d$. For $H = H_0$, the allowed potentials are the other coset $W - H$. For the other two planes containing d , the missing-pair condition leaves M . For each of the remaining four planes it leaves R . This is (4.2).

In the skew case, exactly one plane contains both d, m ; its list is M . Two planes contain d but not m , giving $W - H$. Two contain m but not d ; the initially allowed set R loses exactly one point according as $a + r \in H$. The final two planes contain neither d nor m , and their list is R . This is (4.3) and gives the stated sizes. $\square$

There are $28 \cdot 15 = 420$ ordered disjoint choices of (R, M) : 84 parallel and 336 skew. The sum of either seven-entry profile is 16. Thus the table itself predicts

$$84 \cdot 16 + 336 \cdot 16 = 6,720$$

admitted local triangles among the $420 \cdot 7 = 2,940$ pair-pair-plane cases. The two supplied programs reproduce these counts directly.

5 Compression to five coordinates

Proposition 5.1 (one-hole quotient). *Every solution of Theorem 3.1 yields a standard five-cycle double cover of G .*

Proof. Write $M = \{a, b\}$. Identify a and b , while leaving the other four members of $S = W - R$ distinct. This gives five coordinate names. No used pair collapses to a singleton, because M is the only pair whose endpoints are identified and it is unused.

Before identification, every old coordinate has even degree at each graph vertex. After identification, the degree of the merged coordinate is, modulo two, the sum of the old a - and b -degrees. It is therefore even, as are the four unchanged coordinates. Each edge remains labelled by two distinct new names, so it is covered exactly twice. $\square$

The 14 permitted pair types make the compression explicit. Six pairs avoid both a, b and map injectively to six target pairs. The eight pairs joining one of a, b to one of the other four points map two-to-one to the remaining four target pairs. This is not a fixed pointwise map from the seven nonzero flow values to ten target pairs: two preimages of one target pair have differences separated by $a + b$.

Corollary 5.2 (exact graph-level scope). *A finite loopless cubic multigraph has a standard five-cycle double cover if and only if there exist a nowhere-zero W -flow f and disjoint pairs R, M for which (3.3)–(3.4) is soluble.*

Proof. The reverse implication is Proposition 5.1. Conversely, label each edge of a five-cycle double cover by the two cover indices that contain it. At a cubic vertex every coordinate occurs zero or two times, so the three edge labels form a triangle, as in the proof of Lemma 2.1.

Inject the five coordinate names into any five members of a six-set $S \subset W$, leaving one point $q \in S$ unused. Put $R = W - S$, and let M join q to any used point. No original edge uses M . The xor of the two endpoint names of each edge label is nonzero and sums to zero at every vertex, hence gives a nowhere-zero W -flow. The original local triangles give potentials solving the affine system. $\square$

Remark 5.3. Corollary 5.2 is a reformulation of FiveCDC, not a resolution. The new tractability appears only after f, R, M are fixed. A proof of the conjecture would still have to select a successful triple universally; a counterexample would have to exclude every triple.

6 Dual certificates for a fixed failure

After scalarization, write the fixed system as

$$Ax = b \tag{6.1}$$

over $\mathbb{F}_2$. The variables are the three coordinate bits of every vertex potential. Local rows annihilate the direction space of the appropriate affine list, and two edge rows annihilate each edge value.

Proposition 6.1 (left-kernel certificate). *The fixed system (6.1) is inconsistent if and only if there is a binary row multiplier y such that*

$$y^\top A = 0, \quad y^\top b = 1. \tag{6.2}$$

Proof. If (6.2) holds, xoring the selected equations gives $0 = 1$, so no solution exists. Conversely, row reduction of an inconsistent binary affine system produces a zero coefficient row with right side one. Keeping the same row operations on an identity matrix expresses that row as a multiplier y of the original rows. $\square$

This is a short human-checkable certificate for one failed choice (f, R, M) . It is not a certificate that every choice fails. The distinction is essential: a graph-level counterexample would require a certified universal exclusion over flows as well as pairs.

7 A strict 34-vertex fixed-flow separation

The retained simple connected bridgeless cubic graph has graph6 record

```
as???SK???A?A?C_@_?B?@_??G??_I?GA_AA????_??@?????B??@_?o??????_C??CG0??B@????a????_??C@
0??A?_
```

and 51 edges. Its displayed Fano flow and compatible pair labels are frozen in the project artifact

output/jaeger-star-thinning-countermodel-34v/star-good-six-witness.json

whose SHA-256 is

e3160e5f6b2985823a3421e9f3b1a4582c9c8be4b9abfd1f0e2312d311eb8358.

The literal cover uses the six points $\{0, 3, 4, 5, 6, 7\}$ and all 14 pair types except $\{4, 5\}$. Identifying 4 and 5 gives its displayed FiveCDC.

Proposition 7.1 (verified strict separation). *For this fixed flow:*

1. *none of the 56 five-point restrictions is soluble;*
2. *exactly eight of the 420 six-point one-hole systems are soluble;*
3. *all eight successful choices are skew and have affine solution dimension one.*

This finite proposition is obtained by exact enumeration in two independently structured programs. The primary Python checker uses full row reduction; the JavaScript checker uses an incremental highest-pivot xor basis. Both reconstruct the graph, validate simplicity, connectedness, cubicity, bridgelessness, the flow, and the literal cover. They then test all 56 and 420 choices.

Two five-row dual certificates illustrate that individual failures can also be checked by hand. For the failed five-set $\{0, 1, 2, 3, 4\}$, vertices 0 and 1 both have forced potential 7. At each vertex, take the local rows for functionals 1 and 2; their xor is functional 3 and their two right sides sum to zero. Edge 0 joins the two vertices, has flow value 3, and its functional-3 compatibility row has right side one. Xoring the five rows cancels every variable and yields $0 = 1$.

For the failed six-hole choice $R = \{0, 1\}$, $M = \{2, 3\}$, vertices 4 and 8 both have local list $\{2, 3\}$. At each vertex, xor the functional-4 row of right side zero with the functional-6 row of right side one; the coefficient is functional 2. Edge 5 joins the vertices, has flow value 1, and its functional-2 compatibility row has right side one. Again all coefficients cancel while the five right sides xor to one. The audit emits and rechecks the exact row indices and metadata.

The older five-point restriction has local profile

$$(4, 1, 1, 1, 1, 1, 1).$$

Only one Fano line carries freedom; eliminating forced variables gives the component-parity system of Hušek and Šámal [2]. The profiles in Theorem 4.1 leave freedom on several lines, allowing their global equations to couple. Proposition 7.1 proves that this is a genuine enlargement at fixed flow.

8 A complete all-flow calibration on 12 vertices

For the cubic graph with graph6 record

$$K??FEaKR@oE_,$$

a retained complete census contains 900 $GL(3, 2)$ -orbits of nowhere-zero flows, representing 150,192 labelled flows. Global translation reduces the 56 five-point supports to seven representatives and the 420 pairs (R, M) to 63 representatives.

For comparison with the two restricted tests, call a compatible potential *generally successful* when its coordinate co-occurrence graph on W is properly colourable with at most five colours. Merging coordinates by such a colouring gives a FiveCDC. Conversely, every FiveCDC embeds in this form; this is the unrestricted compatible-potential test for a fixed flow.

Proposition 8.1 (verified all-flow census). *Two exact implementations independently obtain the following partition.*

<i>fixed-flow outcome</i>	<i>flow orbits</i>	<i>labelled flows</i>
<i>five-point soluble</i>	566	94,080
<i>five-point insoluble, six-hole soluble</i>	252	42,336
<i>six-hole insoluble, generally successful</i>	18	3,024
<i>not generally successful</i>	64	10,752

Thus six-point one-hole compression rescues 252 of the 334 flow orbits that fail every five-point restriction. It is strictly broader than the five-point criterion, but not the most general compatible-potential compression: 18 further orbits need a co-occurrence pattern using more than six coordinate points. The 64 remaining flows are obstructions only for those fixed flows, not for the graph.

One rigid flow among the last 64 has edge-value word

$$(1, 2, 3, 4, 7, 3, 5, 3, 6, 5, 7, 2, 5, 1, 4, 6, 2, 4).$$

Its gauged compatible potential is unique and its co-occurrence graph is K_6 , using all 15 pairs on six points. Hence translation cannot create a missing pair. The same graph has other flows that pass the stronger five-point test, demonstrating again why the fixed-flow and graph-level quantifiers must be separated.

9 Reproducibility

The exact artifact package is

`scratch/six-point-one-hole-affine-20260729/`

From the project root, the four principal commands are:

```
python3 scratch/six-point-one-hole-affine-20260729/\
audit_six_point_one_hole_affine.py
node scratch/six-point-one-hole-affine-20260729/\
verify_six_point_one_hole_affine.mjs
python3 scratch/six-point-one-hole-affine-20260729/\
audit_all_flows_12v.py
node scratch/six-point-one-hole-affine-20260729/\
verify_all_flows_12v.mjs
```

The line continuations above are typographical; the repository README gives literal one-line shell commands. All four reports end with status **PASS**. The package’s SHA-256 ledger has digest

`8e89160b7eac0f7aaf5c4fe1b369bddbf89ae7f013211b4022257dfc09258349.`

The first two programs derive the local table, inspect all 2,940 local cases, and replay Proposition 7.1; the latter two independently replay Proposition 8.1. The reference-hash ledger also freezes every external input used by these checks.

The human proof of Theorems 3.1 and 4.1, Proposition 5.1, and Corollary 5.2 is entirely contained in this manuscript. The strict witness and census propositions depend on exact finite enumeration. No SAT solver or uncheckable UNSAT assertion is used in those two propositions.

10 What is new, and what remains open

The conservative possible-novelty claim is the combination of:

1. the six-point one-hole affine normal form for fixed (f, R, M) ;
2. the complete parallel/skew local-list classification;
3. the left-kernel certificate interface for a failed fixed choice;
4. the strict 34-vertex fixed-flow separation from every five-point restriction; and
5. the exact 12-vertex all-flow calibration.

The project has not established priority. The relevant literature on Fano flows, generalized colourings, and the new component-parity criterion includes Jaeger [3], the projective/affine/abelian colouring framework of Král’ et al. [4], Oum [5], and Hušek–Šámal [2]. In particular, no novelty

is claimed for the Desargues-configuration equivalence, F_i/A_i colourings, the eight-coordinate construction, or the five-point component-parity criterion. A specialist literature review is required before asserting novelty in a submitted version.

The universal remaining statement is: for every graph in a sound reduced cubic class, select some nowhere-zero W -flow f and some disjoint R, M for which the affine system is soluble. By Corollary 5.2, this is exactly FiveCDC. Complete finite success of selected flow families is evidence, not an induction theorem. Conversely, failure of one or many fixed systems is not a graph counterexample. No such counterexample, and no universal selection proof, is supplied here.

AI-use disclosure

OpenAI Codex agents using GPT-5-series models, under Atharva Vaidya’s direction, proposed the six-point one-hole compression, derived the affine equations and local orbit tables, found and interpreted the finite examples, wrote the Python and JavaScript implementations, executed the computations, and drafted essentially all prose and L^AT_EX in this manuscript. Separate agents performed adversarial checks, but they were also AI systems. Atharva Vaidya directed the project and is responsible for deciding whether and how to disseminate the work.

The paper deliberately distinguishes human-checkable algebra from finite computation. Reproducibility and independently structured code do not constitute independent human verification. No cited researcher is represented as endorsing these results. Before public submission, a human author should verify the displayed proofs line by line, inspect the graph and flow data, rerun and preferably independently reimplement the enumerations, conduct a specialist prior-art review, and take full responsibility for all claims.

References

- [1] U. A. Celmins. *On Cubic Graphs That Do Not Have an Edge-3-Colouring*. PhD thesis, University of Waterloo, 1984.
- [2] R. Hušek and R. Šámal. Exponentially many circuit double covers. arXiv:2607.24724 [math.CO], version 1, July 27, 2026. <https://arxiv.org/abs/2607.24724>.
- [3] F. Jaeger. Flows and generalized coloring theorems in graphs. *Journal of Combinatorial Theory, Series B*, 26(2):205–216, 1979. [https://doi.org/10.1016/0095-8956\(79\)90057-1](https://doi.org/10.1016/0095-8956(79)90057-1).
- [4] D. Král’, E. Máčajová, O. Pangrác, A. Raspaud, J.-S. Sereni, and M. Škoviera. Projective, affine, and abelian colorings of cubic graphs. *European Journal of Combinatorics*, 30:53–69, 2009. <https://doi.org/10.1016/j.ejc.2007.11.029>.
- [5] S.-i. Oum. A proof of the Cycle Double Cover Conjecture by OpenAI: an exposition. arXiv:2607.16356 [math.CO], version 2, July 23, 2026. <https://arxiv.org/abs/2607.16356>.
- [6] M. Preissmann. *Sur les colorations des arêtes des graphes cubiques*. PhD thesis, Université de Grenoble, 1981.